HELENA M. VARGA

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APPOINTMENTS

2022 – now **Associate Professor of Computational Ecology**, *University of Edinburgh*
Lead of the Landscape Dynamics Group: six doctoral students, two postdoctoral fellows.

2018 – 2022 **Lecturer in Quantitative Ecology**, *University of Edinburgh*

2015 – 2018 **Postdoctoral Research Fellow**, *Swiss Federal Research Institute WSL, Birmensdorf*
Process-based models of alpine treeline shifts under warming; host: Prof. N. Zimmermann.

EDUCATION

2011 – 2015 **PhD in Ecology**, *University of Cambridge*
Thesis: *Seed dispersal and the spatial structure of fragmented woodland*. Advisor: Prof. D. Coomes.

2010 – 2011 **MSc in Applied Statistics, with distinction**, *University of Oxford*

2007 – 2010 **BSc in Biology, first class**, *Eötvös Loránd University, Budapest*

PUBLICATIONS

JOURNAL ARTICLES

J6 **H. M. Varga**, R. Okafor and L. Chen (2025). Early-warning signals of forest dieback from Sentinel-2 time series. *Nature Ecology & Evolution* 9, 412–423.

J5 T. Lindqvist and **H. M. Varga** (2024). Stochastic dispersal kernels recover range-shift lags in European beech. *Ecology Letters* 27, e14318.

J4 **H. M. Varga** and N. E. Zimmermann (2022). Thermal debt at the alpine treeline: a process-based synthesis. *Global Change Biology* 28, 5503–5517.

J3 A. Moreau, **H. M. Varga** and S. Patel (2021). Bayesian occupancy models for camera-trap networks with imperfect detection. *Methods in Ecology and Evolution* 12, 1879–1891.

J2 **H. M. Varga**, D. A. Coomes (2017). Fragment size and the seed shadow of wind-dispersed trees. *Journal of Ecology* 105, 1102–1114.

J1 **H. M. Varga** (2015). Connectivity thresholds in fragmented woodland. *Landscape Ecology* 30, 1745–1758.

CONFERENCE PAPERS

C2 R. Okafor and **H. M. Varga** (2024). Self-supervised canopy segmentation for sparse field labels. *Proc. ICLR Workshop on Tackling Climate Change with Machine Learning*.

C1 **H. M. Varga** and J. Hartmann (2020). Scalable agent-based simulation of seed dispersal on GPUs. *Proc. International Congress on Environmental Modelling and Software*, 211–218.

PREPRINTS

P1 **H. M. Varga**, L. Chen and M. Rossi (2026). A common currency for resilience: comparing recovery rates across biomes. *bioRxiv* 2026.03.11.584102.

GRANTS & AWARDS

2024 – 2029 **ERC Consolidator Grant, DIEBACK**, *€ 1.98 M, principal investigator*

2021 – 2024 **NERC Standard Grant NE/V009214/1**, *£ 612 k, principal investigator*

2023 **British Ecological Society Founders' Prize**

2016 **Marie Skłodowska-Curie Individual Fellowship**, *€ 175 k*

2015 **Harper Prize for the best doctoral thesis**, *British Ecological Society*

INVITED TALKS

2025 **Reading the forest's pulse from orbit**, *Keynote, Ecological Society of America, Baltimore*

2024 **Why range shifts lag behind climate**, *Max Planck Institute for Biogeochemistry, Jena*

2023 **Models that ecologists can argue with**, *Department of Zoology, University of Oxford*

2022 **Seed shadows in a warming world**, *Swiss Ecology Meeting, Lausanne*

2021 **Occupancy at scale**, *Plenary, EURING Analytical Meeting, online*

2019 **Treelines as natural experiments**, *Department of Plant Sciences, University of Cambridge*

SOFTWARE & DATA

2021 – now **dispersr**, *R package for simulating seed dispersal kernels on real landscapes, on CRAN*
About 4 000 downloads a month; used in courses at five universities.

2024 **Scottish Woodland Dieback Atlas**, *open data set of 1.2 M annotated canopy tiles, Zenodo*

TEACHING

2019 – now **Ecological Modelling**, *honours course, 60 students per year, course organiser*

2018 – now **Statistics for the Life Sciences**, *second-year core course, co-lecturer*

2022 – now **Doctoral supervision**, *four students graduated, six current*

SERVICE

2023 – now **Associate Editor**, *Methods in Ecology and Evolution*

2022 – now **Panel member**, *NERC Discovery Science, Peer Review College*

2020 – 2023 **Equality, Diversity and Inclusion Committee**, *School of Biological Sciences*

2019 – 2021 **Co-organiser**, *Symposium on Spatial Ecology, British Ecological Society Annual Meeting*

Reviewer, *Nature, Ecology Letters, Global Change Biology, PNAS, Remote Sensing of Environment*

REFERENCES

Available on request.